INDUSTRIAL AGENTS HACKATHON 2026 · TRACK 01 · TECHNICAL EXPLANATION

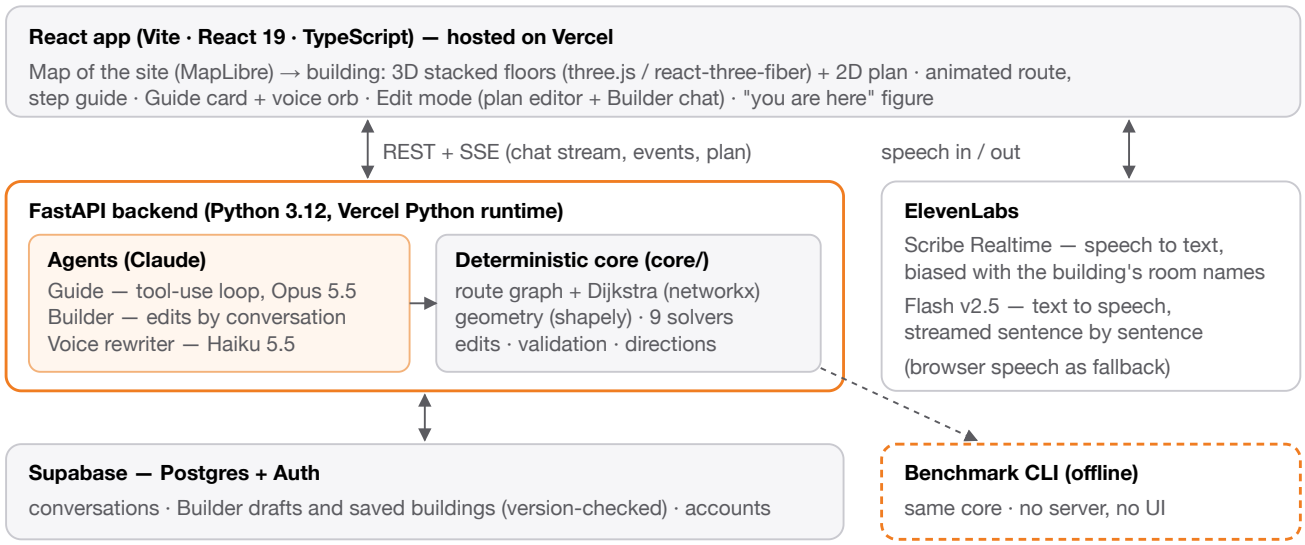
Talk to Your Building: architecture and tech stack

An app you talk to — by text or voice — that finds things, plans routes, reacts to closures and edits the building. A deterministic geometry/graph core computes every answer; Claude handles the language.

> **One rule shapes the whole system: the LLM for language, code for facts.** No room, distance, count or route reaches the user unless it came out of a deterministic tool. The rule is enforced in code (a grounding check after every agent turn), not only in the prompt.

138 / 138	1,338 / 1,338	145 / 145	280
dev benchmark questions correct (official scorer)	train + dev routes reproduced exactly (length, path)	test questions answered, 0 errors (`SUBMISSION/answers.json`)	kit buildings that load, route and render in the app

Architecture



One Python core serves three consumers: the conversational Guide, the Builder and the one-command benchmark.

1 · Deterministic core (`core/`)

- **Route graph.** Nodes are room anchors and doors; edges are walks inside a room (a visibility graph that keeps 15 cm off the walls and goes round corners), stairs (2 m per metre climbed) and lifts (15 m per ride + 1 m per metre climbed). Dijkstra over a `networkx` multigraph. This is the organisers' kit 1.1.0 route model, ported unchanged so our lengths and paths match theirs to the centimetre.
- **Constraints as graph filters.** Step-free drops stairs and stepped doors; closures remove a door, lift, stair or room; a minimum door width (trolley, wheelchair, forklift) drops narrower openings. Routing profiles change these parameters, never the algorithm.
- **Solvers** for every benchmark kind (largest room, floor area, shared walls, stacked rooms, 3D straight line, errands, "what if this door closes", wide deliveries, step-free reachability) plus locate, count, route, closures and multi-stop tours.
- **Directions** a person can follow: turns at the door you came through, a landmark item near each door ("past the printers"), rounded metres, one line per lift or stair ride.
- **Edits and validation** (`core/edits.py`, `core/validate.py`): add/move/delete rooms, doors and equipment, move a shared wall without breaking the neighbour, then check the result.

2 · Agents (`agent/`) — Claude through the Anthropic API

Agent	What it does	Guardrails
Building Guide	Conversation in text or voice: find, route, react to closures, change the building. Manual tool-use loop over 16 tools (`search`, `find_route`, `tour`, `compare_routes`, `set_profile`, `propose_change`, `show_on_map`, …). Claude Opus 5.5; system prompt + building overview prompt-cached.	Grounding check: every number or id in a reply must appear in a tool result, else one repair round. A change proposed in a turn cannot be applied in that same turn — the user sees the effect and confirms.
Builder	Edits a building by conversation ("make the kitchen 2 m wider", "add a door between …"): inspects the plan with read-only tools, proposes an edit with its predicted effect and new problems.	Proposals are dry runs; only the user's next "yes" or the Apply button commits; stale proposals are refused (409).
Benchmark pipeline	Question text → template parser (LLM fallback only for unknown wording) → solver → reply with a deterministic `Answer:` line.	The LLM only maps wording to a kind and ids; every number comes from a solver.

How it answers the four questions

Brief	Example	What happens
FIND	"Where is the 3D printer?"	`search` resolves names, equipment types and typos; several matches → one short question naming what tells them apart (floor, size, neighbour); the room and item are tinted, labelled and framed on the plan.
ROUTE	"From the lobby to the CNC mill on floor 2 without stairs"	The profile becomes a graph filter; the Guide echoes its understanding in one line (from → to, constraints), gives the total and the first step; the app draws the whole route, then guides step by step with "I'm there".
EDIT	"Add a door between rooms 2.03 and 2.04"	`propose_change` → effect card ("I understood: …; these routes change") → confirm → applied and validated; Edit mode shows the change on the plan.
REACT	"The east lift is closed. What changes?"	`compare_routes` computes rooms that lost step-free access (by floor) and routes that got longer, with before/after metres; affected rooms are highlighted.

Built for the people in the brief

Who	What they get
First day at work	Needs no room names: "where do I get coffee?" is resolved through room and equipment types; the answer is drawn on the plan and the app guides step by step from the "you are here" figure.
Wheelchair user	Says it once: the profile (step-free, minimum door width) applies to every later route without asking twice; closures show which rooms lost step-free access.
Service technician	Hands-free voice with barge-in; directions by landmarks ("past the press"), not room ids.
Facility manager	Closes a lift, adds a door or moves a machine by conversation or in Edit mode; sees the effect before confirming.
Forklift driver / logistics	Forklift profile (openings ≥ 3 m, no stairs, no pedestrian walkways); blocked aisles as closures; "where is the steel drum" through the equipment search; moving objects report their new place.

Levels covered

- **1 Locate / 2 Route:** find, step-free routes, closures, minimum door widths, multi-stop tours.
- **3 Scale:** multi-floor routing over stairs and lifts in all 280 kit buildings (up to 10 floors); places in nearby located buildings are searchable and routed to with an outdoor leg.
- **4 Build:** Edit mode — draw and drag rooms, walls, doors and equipment, or ask the Builder in plain language; drafts validated, undoable, saved and immediately routable by the Guide.

- **5 Extend:** voice in and out with barge-in; isometric stacked floors over a city map; accessibility profiles (step-free, wide doors, quiet routes); industrial logistics (forklift routing); moving objects (devices report "this is now here"); per-user access to restricted rooms.

Voice and interaction design

- **Hands-free voice.** Speech goes to ElevenLabs Scribe Realtime with the building's room names as keyterms; a small Haiku rewriter fixes transcription slips before the Guide. Replies are streamed as grounded sentences to ElevenLabs TTS, with a fact-free "let me check" first (first sound ≈ 2.3 s); the user can interrupt at any time.
- **Ask only what changes the answer.** For an ambiguous place the Guide runs the solver for each candidate; if the answers agree it just answers, otherwise it asks one short question with ≤ 3 options.
- **Profile once, remember always.** "I use a wheelchair" sets a profile used by every later route; a deterministic cue detector backs the model up if it forgets.
- **Measured.** An interaction eval has Claude play six tricky users (first day, wheelchair, facility manager, voice technician, typo + change of mind, Turkish speaker) and a judge score the brief's rubric 1–5: asks only what it needs 4.67, shows what it understood 4.83, directions 4.17, remembers 5.00.

Sunday benchmark

One command from the repo root, no manual steps; documented in `SUBMISSION/BENCHMARK.md`:

```
uv sync
uv run --env-file .env python -m agent.run_benchmark_v1 <building_dir> <questions>
SUBMISSION/answers.json
```

Each answer carries the reply, the located room/item id, and for routes the ordered `space_path` / `element_path` and length in metres. Without an API key (`PARSER_MODE=regex`) the run is fully deterministic and offline. Nothing is tuned to a building or question id: the same parser and solvers run on any building in the kit format. Scores with the official `benchmark/score.py`: dev 138/138, train 515/517 (the two misses name a room whose name several rooms share); the 145 test questions are answered by the same command.

Tech stack

Layer	Technology
Core	Python 3.12, uv, shapely (geometry), networkx (graph + Dijkstra), numpy
Backend	FastAPI + uvicorn, Server-Sent Events, OpenAI-compatible `/v1/chat/completions`; Vercel Python runtime
AI	Anthropic API — Claude Opus 5.5 (Guide, Builder, parser fallback), Haiku 5.5 (voice rewriter); prompt caching, tool use, structured outputs
Voice	ElevenLabs Scribe Realtime (STT) and Flash v2.5 (TTS); Web Speech API as fallback
Frontend	React 19, TypeScript, Vite, three.js + react-three-fiber + drei, MapLibre GL, lucide icons
Data	Supabase Postgres + Auth (conversations, drafts, saved buildings, accounts); kit dataset in `data/`
Quality	pytest suite (solvers vs. all train/dev answers, routes, parser, Guide conversations, edits, API), scripted live Guide and interaction evals; every commit checked by an independent verifier agent

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Repository: github.com/atahankilc/industrial-agents-hackathon · Our code MIT; kit code and data keep their own licences (NOTICE.md); building data derived from OpenStreetMap © OpenStreetMap contributors, ODbL.